```
package com.navox.agent;

import android.Manifest;
import android.app.Activity;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.content.pm.ResolveInfo;
import android.graphics.Color;
import android.net.Uri;
import android.os.Build;
import android.os.Bundle;
import android.os.PowerManager;
import android.speech.RecognizerIntent;
import android.app.SearchManager;
import android.content.Context;
import
android.hardware.camera2.CameraCharacterist
ics;
import
android.hardware.camera2.CameraManager;
import android.provider.AlarmClock;
import android.provider.MediaStore;
import android.provider.Settings;
import android.view.View;
import android.webkit.JavascriptInterface;
import android.webkit.WebResourceRequest;
import android.webkit.WebSettings;
import android.webkit.WebView;
import android.webkit.WebViewClient;
import org.json.JSONArray;
```

```
import org.json.JSONObject;
import java.util.ArrayList;

public class MainActivity extends Activity
{
    WebView web;
    String pendingCmd;
    boolean loaded = false;
    static final int VOICE = 1;

    @Override
    protected void onCreate(Bundle b) {
        super.onCreate(b);

getWindow().setStatusBarColor(Color.parseCo
lor("#FCFBF8"));

getWindow().setNavigationBarColor(Color.par
seColor("#FCFBF8"));
        int uiFlags =
View.SYSTEM_UI_FLAG_LIGHT_STATUS_BAR;
        if (Build.VERSION.SDK_INT >= 26)
uiFlags |=
View.SYSTEM_UI_FLAG_LIGHT_NAVIGATION_BAR;

getWindow().getDecorView().setSystemUiVisib
ility(uiFlags);
        web = new WebView(this);

web.setBackgroundColor(Color.parseColor("#F
CFBF8"));
        setContentView(web);
        WebSettings s = web.getSettings();
        s.setJavaScriptEnabled(true);
```

```
        s.setDomStorageEnabled(true);
        s.setAllowFileAccess(true);

s.setAllowUniversalAccessFromFileURLs(true)
;

s.setMediaPlaybackRequiresUserGesture(false
);
        web.addJavascriptInterface(new
Bridge(), "NAVOX");
        web.setWebViewClient(new
WebViewClient() {
            @Override
            public boolean
shouldOverrideUrlLoading(WebView v,
WebResourceRequest r) {
                String u =
r.getUrl().toString();
                if (u.startsWith("file:"))
return false;
                openUrl(u, "");
                return true;
            }

            @Override
            public void
onPageFinished(WebView v, String url) {
                loaded = true;
                deliverCmd();
            }
        });
        pendingCmd =
getIntent().getStringExtra("cmd");
        new Thread(new Runnable() {
```

```
            public void run() {

CommandRunner.loadApps(MainActivity.this);
            }
        }).start();
        if (getSharedPreferences("navox",
0).getBoolean("wake", false) &&
!ListenService.running) {
            new Bridge().startListening();
        }

web.loadUrl("file:///android_asset/index.ht
ml");
    }

    boolean openUrl(String url, String pkg)
{
        boolean hasPkg = pkg != null &&
!pkg.isEmpty();
        try {
            Intent i = new
Intent(Intent.ACTION_VIEW, Uri.parse(url));
            if (hasPkg) i.setPackage(pkg);

i.addFlags(Intent.FLAG_ACTIVITY_NEW_TASK);
            startActivity(i);
            return true;
        } catch (Exception e) {
            return hasPkg && openUrl(url,
"");
        }
    }

    boolean start(Intent i) {
```

```
        try {

i.addFlags(Intent.FLAG_ACTIVITY_NEW_TASK);
            startActivity(i);
            return true;
        } catch (Exception e) {
            return false;
        }
    }

    void js(String code) {
        web.evaluateJavascript(code, null);
    }

    void deliverCmd() {
        if (loaded && pendingCmd != null) {

js("window.onWakeCommand&&window.onWakeComm
and(" + JSONObject.quote(pendingCmd) +
")");
            pendingCmd = null;
        }
    }

    @Override
    protected void onNewIntent(Intent i) {
        super.onNewIntent(i);
        setIntent(i);
        String c = i.getStringExtra("cmd");
        if (c != null) {
            pendingCmd = c;
            deliverCmd();
        }
    }
```

```
    @Override
    protected void onResume() {
        super.onResume();

js("window.onNativeResume&&window.onNativeR
esume()");
    }

    @Override
    public void
onRequestPermissionsResult(int rq, String[]
p, int[] r) {

super.onRequestPermissionsResult(rq, p, r);

js("window.onNativeResume&&window.onNativeR
esume()");
    }

    class Bridge {
        @JavascriptInterface
        public boolean launch(String pkg) {
            try {
                PackageManager pm =
getPackageManager();
                Intent i =
pm.getLaunchIntentForPackage(pkg);
                if (i == null) {
                    Intent q = new
Intent(Intent.ACTION_MAIN);

q.addCategory(Intent.CATEGORY_LAUNCHER);
                    q.setPackage(pkg);
```

```
java.util.List<ResolveInfo> l =
pm.queryIntentActivities(q, 0);
                    if (l.isEmpty()) return
false;
                    i = new
Intent(Intent.ACTION_MAIN);

i.addCategory(Intent.CATEGORY_LAUNCHER);
                    i.setClassName(pkg,
l.get(0).activityInfo.name);
                }

i.addFlags(Intent.FLAG_ACTIVITY_NEW_TASK);
                startActivity(i);
                return true;
            } catch (Exception e) {
                return false;
            }
        }

        @JavascriptInterface
        public boolean openUrl(String url,
String pkg) {
            return
MainActivity.this.openUrl(url, pkg);
        }

        @JavascriptInterface
        public boolean dial(String n) {
            return start(new
Intent(Intent.ACTION_DIAL, Uri.parse("tel:"
+ Uri.encode(n))));
        }
```

```
        @JavascriptInterface
        public boolean sms(String n, String
body) {
            Intent i = new
Intent(Intent.ACTION_SENDTO,
Uri.parse("smsto:" + Uri.encode(n)));
            i.putExtra("sms_body", body);
            return start(i);
        }

        @JavascriptInterface
        public boolean alarm(int h, int m,
String label) {
            Intent i = new
Intent(AlarmClock.ACTION_SET_ALARM);

i.putExtra(AlarmClock.EXTRA_HOUR, h);

i.putExtra(AlarmClock.EXTRA_MINUTES, m);

i.putExtra(AlarmClock.EXTRA_MESSAGE,
label);

i.putExtra(AlarmClock.EXTRA_SKIP_UI, true);
            return start(i);
        }

        @JavascriptInterface
        public boolean timer(int sec,
String label) {
            Intent i = new
Intent(AlarmClock.ACTION_SET_TIMER);
```

```
i.putExtra(AlarmClock.EXTRA_LENGTH, sec);

i.putExtra(AlarmClock.EXTRA_MESSAGE,
label);

i.putExtra(AlarmClock.EXTRA_SKIP_UI, true);
            return start(i);
        }

        @JavascriptInterface
        public boolean settings(String w) {
            String a =
Settings.ACTION_SETTINGS;
            if (w.equals("wifi")) a =
Settings.ACTION_WIFI_SETTINGS;
            else if (w.equals("bluetooth"))
a = Settings.ACTION_BLUETOOTH_SETTINGS;
            else if (w.equals("location"))
a =
Settings.ACTION_LOCATION_SOURCE_SETTINGS;
            else if (w.equals("display")) a
= Settings.ACTION_DISPLAY_SETTINGS;
            else if (w.equals("sound")) a =
Settings.ACTION_SOUND_SETTINGS;
            else if (w.equals("airplane"))
a = Settings.ACTION_AIRPLANE_MODE_SETTINGS;
            else if (w.equals("battery")) a
= Settings.ACTION_BATTERY_SAVER_SETTINGS;
            return start(new Intent(a));
        }

        @JavascriptInterface
        public boolean play(String q) {
            Intent i = new
```

```
Intent(MediaStore.INTENT_ACTION_MEDIA_PLAY_
FROM_SEARCH);
            i.putExtra(SearchManager.QUERY,
q);

i.putExtra(MediaStore.EXTRA_MEDIA_FOCUS,
"vnd.android.cursor.item/*");
            return start(i);
        }

        @JavascriptInterface
        public boolean torch(boolean on) {
            try {
                CameraManager cm =
(CameraManager)
getSystemService(Context.CAMERA_SERVICE);
                for (String id :
cm.getCameraIdList()) {
                    Boolean f =
cm.getCameraCharacteristics(id).get(CameraC
haracteristics.FLASH_INFO_AVAILABLE);
                    if (f != null && f) {
                        cm.setTorchMode(id,
on);
                        return true;
                    }
                }
            } catch (Exception e) { }
            return false;
        }

        @JavascriptInterface
        public String setupStatus() {
            JSONObject o = new
```

```
JSONObject();
            try {
                o.put("mic",
checkSelfPermission(Manifest.permission.REC
ORD_AUDIO) ==
PackageManager.PERMISSION_GRANTED);
                o.put("notif",
Build.VERSION.SDK_INT < 33 ||
checkSelfPermission("android.permission.POS
T_NOTIFICATIONS") ==
PackageManager.PERMISSION_GRANTED);
                PowerManager pm =
(PowerManager)
getSystemService(Context.POWER_SERVICE);
                o.put("battery",
pm.isIgnoringBatteryOptimizations(getPackag
eName()));
                o.put("overlay",
Settings.canDrawOverlays(MainActivity.this)
);
                o.put("listening",
ListenService.running);
                o.put("access",
NavoxAccess.inst != null);
            } catch (Exception e) { }
            return o.toString();
        }

        @JavascriptInterface
        public void askMic() {
            runOnUiThread(new Runnable() {
                public void run() {
                    requestPermissions(new
String[]{Manifest.permission.RECORD_AUDIO},
```

```
2);
                }
            });
        }

        @JavascriptInterface
        public void askNotif() {
            runOnUiThread(new Runnable() {
                public void run() {
                    if
(Build.VERSION.SDK_INT >= 33)
requestPermissions(new String[]
{"android.permission.POST_NOTIFICATIONS"},
3);
                }
            });
        }

        @JavascriptInterface
        public void askBattery() {
            runOnUiThread(new Runnable() {
                public void run() {
                    start(new
Intent(Settings.ACTION_REQUEST_IGNORE_BATTE
RY_OPTIMIZATIONS, Uri.parse("package:" +
getPackageName())));
                }
            });
        }

        @JavascriptInterface
        public void askOverlay() {
            runOnUiThread(new Runnable() {
                public void run() {
```

```
                    start(new
Intent(Settings.ACTION_MANAGE_OVERLAY_PERMI
SSION, Uri.parse("package:" +
getPackageName())));
                }
            });
        }

        @JavascriptInterface
        public void setLang(String l) {
            getSharedPreferences("navox",
0).edit().putString("lang", l).apply();
        }

        @JavascriptInterface
        public boolean startListening() {
            try {
                if
(checkSelfPermission(Manifest.permission.RE
CORD_AUDIO) !=
PackageManager.PERMISSION_GRANTED) return
false;

getSharedPreferences("navox",
0).edit().putBoolean("wake", true).apply();
                Intent s = new
Intent(MainActivity.this,
ListenService.class);
                if (Build.VERSION.SDK_INT
>= 26) startForegroundService(s); else
startService(s);
                return true;
            } catch (Exception e) {
                return false;
```

```
            }
        }

        @JavascriptInterface
        public boolean stopListening() {
            getSharedPreferences("navox",
0).edit().putBoolean("wake",
false).apply();
            return stopService(new
Intent(MainActivity.this,
ListenService.class));
        }

        @JavascriptInterface
        public String exec(String text) {
            String r =
CommandRunner.run(MainActivity.this, text);
            return r == null ? "" : r;
        }

        @JavascriptInterface
        public void askAccess() {
            runOnUiThread(new Runnable() {
                public void run() {
                    start(new
Intent(Settings.ACTION_ACCESSIBILITY_SETTIN
GS));
                }
            });
        }

        @JavascriptInterface
        public String getApps() {
            JSONArray a = new JSONArray();
```

```
            try {
                PackageManager pm =
getPackageManager();
                Intent q = new
Intent(Intent.ACTION_MAIN);

q.addCategory(Intent.CATEGORY_LAUNCHER);
                for (ResolveInfo r :
pm.queryIntentActivities(q, 0)) {
                    JSONObject o = new
JSONObject();
                    o.put("n",
r.loadLabel(pm).toString());
                    o.put("p",
r.activityInfo.packageName);
                    a.put(o);
                }
            } catch (Exception e) { }
            return a.toString();
        }

        @JavascriptInterface
        public void listen(final String
lang) {
            runOnUiThread(new Runnable() {
                public void run() {
                    try {
                        Intent i = new
Intent(RecognizerIntent.ACTION_RECOGNIZE_SP
EECH);

i.putExtra(RecognizerIntent.EXTRA_LANGUAGE_
MODEL,
RecognizerIntent.LANGUAGE_MODEL_FREE_FORM);
```

```
i.putExtra(RecognizerIntent.EXTRA_LANGUAGE,
lang);

startActivityForResult(i, VOICE);
                    } catch (Exception e) {

js("window.onNativeVoiceError&&window.onNat
iveVoiceError()");
                    }
                }
            });
        }
    }

    @Override
    protected void onActivityResult(int rq,
int rs, Intent d) {
        super.onActivityResult(rq, rs, d);
        if (rq != VOICE) return;
        if (rs == RESULT_OK && d != null) {
            ArrayList<String> l =
d.getStringArrayListExtra(RecognizerIntent.
EXTRA_RESULTS);
            if (l != null && !l.isEmpty())
{
                js("window.onNativeVoice("
+ JSONObject.quote(l.get(0)) + ")");
                return;
            }
        }

js("window.onNativeVoiceError&&window.onNat
iveVoiceError()");
```

```
    }

    @Override
    public void onBackPressed() {
        if (web.canGoBack()) web.goBack();
else super.onBackPressed();
    }
}
```